

Mathematical induction

Base case · inductive hypothesis · inductive step · strong induction

Topics: ordinary induction, strong induction, sums, divisibility, and common errors.

Today's goal: prove a claim for every whole number from a starting value onward, without checking the numbers one by one.

Session	Date
6 of 15	Mon 14 Sep 2026

Where session 6 sits

1 · Course overview and school-maths refresher

2 · Sets and set operations

3 · Functions

4 · Function families

5 · Theorem proving

6 · Mathematical induction

7 · Elementary combinatorics and binomial theorem

8 · Midterm

9 · GCD and divisibility

10 · Modular arithmetic

11 · Vectors

12 · Polynomials

13 · Complex numbers

14 · Course recap and exam preparation

15 · Final exam

TODAY Ordinary induction, strong induction, and how to spot a broken induction proof.

Yesterday's proof methods, today's new question

In Session 5, we learned how to prove one claim or disprove it with a counterexample.

Today, suppose we want to prove a statement for

$$n = 1, 2, 3, 4, \dots$$

Today's question

How can we prove all of these infinitely many cases without writing infinitely many separate proofs?

A pattern suggests a formula; it does not prove it

CONSIDER THE CLAIM

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

n	LEFT SIDE	RIGHT SIDE
1	1	1
2	3	3
3	6	6
4	10	10

NOT YET A PROOF The examples make the formula believable.
They do **not** prove it for every positive integer n .

The domino picture

ANALOGY Imagine an endless line of dominoes numbered $1, 2, 3, \dots$

1. Show that domino 1 falls.
2. Show that whenever domino k falls, domino $k + 1$ also falls.

Then every domino falls.

TRANSLATION In induction, "domino n falls" becomes "the statement $P(n)$ is true."

CAREFUL The dominoes are only an analogy. A mathematical proof must still show the base case and the step from k to $k + 1$.

The principle of mathematical induction

DEFINITION Let $P(n)$ mean a statement about an integer n , starting at a whole number n_0 .

THE RULE

1. **Base case:** show that $P(n_0)$ is true.
2. **Inductive step:** take any integer $k \geq n_0$. Assume $P(k)$ is true. Use this to show that $P(k + 1)$ is true.
3. **Conclusion:** $P(n)$ is true for every integer $n \geq n_0$.

STATUS This is the induction rule. It is a basic property of the whole numbers.

First name the statement $P(n)$

THE STATEMENT We will prove, for every integer $n \geq 1$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Let $P(n)$ mean this exact statement.

EXAMPLE For example, $P(4)$ means

$$1 + 2 + 3 + 4 = \frac{4(4+1)}{2}.$$

WHY LABEL IT The label $P(n)$ tells us exactly what we assume and exactly what we must prove next.

Base case: start at the stated first value

BASE CASE For the sum formula, the claim starts at $n = 1$.

$$P(1) : \quad 1 = \frac{1(1+1)}{2} = 1.$$

So $P(1)$ is true.

START WHERE THE CLAIM STARTS This starts the chain. A correct calculation at $n = 2$ would not replace the required base case at $n = 1$.

Inductive hypothesis: one temporary assumption

INDUCTIVE HYPOTHESIS Let $k \geq 1$ be **any** integer. Assume that $P(k)$ is true:

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}.$$

This is the **inductive hypothesis**.

ALLOWED We may use it in this step to prove $P(k+1)$.

NOT ALLOWED We are **not** assuming the formula is already true for every n .

Inductive step: use the hypothesis once

STEP Starting with the left side of $P(k+1)$,

$$\begin{aligned} 1 + 2 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \quad \text{by } P(k) \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

This is exactly the right side of $P(k+1)$. Therefore $P(k) \Rightarrow P(k+1)$.

THE DECISIVE LINE The line marked "by $P(k)$ " is the line that makes this an induction proof.

Finish the proof with the induction principle

WE HAVE SHOWN

- $P(1)$ is true.
- For every integer $k \geq 1$, $P(k) \Rightarrow P(k + 1)$.

CONCLUSION

Therefore, by mathematical induction,

$$1 + 2 + \cdots + n = \frac{n(n + 1)}{2}$$

for every integer $n \geq 1$. ■

Build an induction proof

ACTIVITY 1 PAIRS

12 MIN

Prove, for every integer $n \geq 1$,

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2.$$

1. Write $P(n)$ in words or symbols.
2. Check the base case.
3. State the inductive hypothesis.
4. Add the next odd number and prove $P(k + 1)$.

A complete walkthrough

METHOD Let $P(n)$ be the displayed statement.

- **Base case:** $P(1)$ says $1 = 1^2$, which is true.
- **Hypothesis:** assume $1 + 3 + \cdots + (2k - 1) = k^2$ for some $k \geq 1$.

STEP

$$\begin{aligned} 1 + 3 + \cdots + (2k - 1) + (2k + 1) &= k^2 + 2k + 1 \\ &= (k + 1)^2. \end{aligned}$$

So $P(k + 1)$ is true. Induction proves $P(n)$ for every $n \geq 1$. ■

SECTION 2

Induction can prove more than sum formulas

The same structure works for multiples and for statements that need an earlier case other than $P(k)$.

Next, we will prove a divisibility statement, then learn **strong induction**.



Make the required multiple appear

CLAIM For every integer $n \geq 1$, $n^3 - n$ is a multiple of 3.

THE WORDS, IN ALGEBRA "A multiple of 3" means "3 times an integer."

BASE CASE

$$1^3 - 1 = 0 = 3 \cdot 0.$$

So the claim is true when $n = 1$.

The inductive step

STEP Take any integer $k \geq 1$. Assume the claim is true for k . Then $k^3 - k = 3m$ for some integer m . Therefore

$$\begin{aligned}(k+1)^3 - (k+1) &= (k^3 - k) + 3k(k+1) \\ &= 3m + 3k(k+1) \\ &= 3(m + k(k+1)).\end{aligned}$$

The number in brackets is an integer. So the claim is true for $k+1$. By induction, 3 divides $n^3 - n$ for every $n \geq 1$. ■

TWO DIFFERENT LINES The first line is an algebra identity. The second line is where the inductive hypothesis is used.

Find the first wrong line

ACTIVITY 2 GROUPS OF THREE

10 MIN

Someone claims that, for every $n \geq 1$,

$$1 + 2 + \cdots + n = n^2.$$

Their base case $n = 1$ is correct. They assume the claim for k and then write

$$1 + 2 + \cdots + k + (k + 1) = k^2 + (k + 1) = (k + 1)^2.$$

1. Find the first false equality.
2. Explain why the base case does not save the proof.
3. State the correct formula for $1 + 2 + \cdots + n$.

Diagnose and repair

METHOD The first false equality is

$$k^2 + k + 1 = (k + 1)^2.$$

But

$$(k + 1)^2 = k^2 + 2k + 1,$$

so the two expressions differ by k .

THE TRAP The claimed formula is already false at $n = 2$:

$$1 + 2 = 3 \neq 4 = 2^2.$$

THE CORRECT FORMULA

$$1 + 2 + \cdots + n = \frac{n(n + 1)}{2}.$$

Use all earlier cases when needed

DEFINITION Ordinary induction assumes $P(k)$ to prove $P(k + 1)$.

Strong induction assumes the statement is true for every integer from n_0 to k :

$$P(n_0), P(n_0 + 1), \dots, P(k)$$

to prove $P(k + 1)$.

SAME CONCLUSION The conclusion is the same: $P(n)$ is true for every $n \geq n_0$.

WHEN TO USE IT Use strong induction when the next case needs an earlier result, such as $P(k - 3)$, not only $P(k)$.

Prime factorisation: the statement and its words

DEFINITION A positive integer greater than 1 is **prime** if its only positive factors are 1 and itself.

DEFINITION It is **composite** if it can be written as

$$n = ab \quad \text{with} \quad 1 < a, b < n.$$

CLAIM Every integer $n \geq 2$ is a product of prime numbers.
A prime number itself counts as a product with one factor.

EXAMPLE 13 is prime, while $12 = 3 \cdot 4$ is composite.

Strong-induction proof · prime factorisation

PROOF **Base case:** 2 is prime, so it is a product with one prime factor.

Take any integer $k \geq 2$. Assume every integer from 2 to k is a product of primes. Consider $k + 1$.

- If $k + 1$ is prime, we are done.
- If $k + 1$ is composite, write $k + 1 = ab$ with $2 \leq a, b \leq k$.

By the hypothesis, both a and b are products of primes.

Therefore $ab = k + 1$ is a product of primes. Strong induction proves the claim for every integer $n \geq 2$. ■

WHERE THE STRENGTH IS USED The factors a and b can be anywhere between 2 and k , so $P(k)$ alone is not enough.

When strong induction is useful

ACTIVITY 3 PAIRS

14 MIN

Prove that every integer $n \geq 12$ can be written as

$$n = 4a + 5b$$

for non-negative integers a and b . Here, non-negative means $0, 1, 2, \dots$

1. Check the four base cases $12, 13, 14, 15$.
2. State a strong inductive hypothesis.
3. For $k \geq 15$, make $k + 1$ by adding 4 to the earlier amount $k - 3$.

A strong-induction walkthrough

METHOD The four base cases are

$$12 = 3 \cdot 4, \quad 13 = 2 \cdot 4 + 5,$$

$$14 = 4 + 2 \cdot 5, \quad 15 = 3 \cdot 5.$$

Take any integer $k \geq 15$. Assume every integer from 12 to k has the form $4a + 5b$.

STEP Since $12 \leq k - 3 \leq k$, write $k - 3 = 4a + 5b$. Then

$$k + 1 = (k - 3) + 4 = 4(a + 1) + 5b.$$

So $k + 1$ also has the required form. Strong induction proves the claim for all $n \geq 12$. ■

Five induction mistakes to catch early

MISTAKE

1. Starting at the wrong value.
2. Hiding the inductive hypothesis.
3. Using the hypothesis for every n .
4. Never using the hypothesis.
5. Proving the wrong next statement.

FIX

1. Prove the base case at the first stated value n_0 .
2. Write "assume $P(k)$ " or state the equation you may use.
3. Assume it only for one allowed integer k , chosen freely.
4. Mark the line where $P(k)$ is used.
5. Check that the final line is exactly $P(k + 1)$.

Work independently

1 For $1 + 3 + \cdots + (2n - 1) = n^2$, write the base case and the inductive hypothesis.

2 Complete the step: if $2^k \geq k + 1$ for $k \geq 0$, explain why $2^{k+1} = 2 \cdot 2^k \geq 2(k + 1) \geq k + 2$.

3 In the postage activity, which earlier statement is used to prove $P(k + 1)$?

RULE Write one sentence explaining why a base case alone is not an induction proof.

What today was about

- **Induction** proves a statement for every integer from a stated starting value onward.
- A complete proof needs a **base case**, an **inductive hypothesis**, and an **inductive step**.
- **Strong induction** lets the next case use any earlier case in the stated range.
- A proof that skips a base case, hides its hypothesis, or makes a false algebra step is not complete.

base case

inductive hypothesis

inductive step

strong induction

Problem Set 6

Six induction proofs, including one strong-induction proof, plus one broken proof to diagnose.

Next session: **elementary combinatorics and the binomial theorem**.